MILITARY SPECIFICATION

PROPELLANT, ROCKET MOTOR: ROCKET, HIGH EXPLOSIVE, 345 MILLIMETER, XM 130

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for the formulation of one type of propellant for the rocket motor for the Rocket, High, Explosive, 345 Millimeter: XM 130, herein referred as the propellant.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-M-261	- Methyl Ethyl Ketone, Technical.

MILITARY

MIL-A-192	- Ammonium Perchlorate (Technical).
MIL-P-14067	- Powder, Metal, Atomized (for use in Ammunition).
MIL-F-23938	- Ferric Oxide for Propellant Grains Mark 75 and Mark 76.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AS-3892	- Polybutadiene, Liquid Hydroxyl Terminated.
XAS-4600	- Curing Agent, Isophorone Diisocyanate (IPDI).
XAS-4867	- Antioxidant Compound, Propellant.

Naval Sea Systems Command
(Code Ident 53711)

WS-7692	- n-Hexane.

(Copies of specifications, standards, handbooks, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting office.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D792 - Specific Gravity and Density of Plastics by Displacement, Standard Methods of Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1901 Race Street, Philadelphia, Pennsylvania 19103.)

CHEMICAL PROPULSION INFORMATION AGENCY (CPIA)

CPIA Publication No. 21 - CPIA Solid Propellant Mechanical Behavior Manual.

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(Application for copies should be addressed to the Chemical Propulsion Information Agency, The Johns Hopkins University, Applied Physics Laboratory, Johns Hopkins Road, Laurel, Maryland 20810.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General material requirements. The propellant shall meet the following product characteristics and quality.

3.1.1 Formulation. The propellant shall be formulated in accordance with table I and as specified herein.

TABLE I. Propellant formulation.

Ingredient	Applicable specification	Percent by weight
Ammonium perchlorate	MIL-A-192, grade A, class 4	66.000 1/
Aluminum powder	MIL-P-14067 type 2	12.000
Aluminum oxide 2/	-	5.000
Iron oxide (yellow) 3/	MIL-F-23938	1.000
Liquid HTPB, Type 1	AS-3892	(see 3.1.1.2)
Antioxidant compound propellant	XAS-4867	(see 3.1.1.2)
Curing agent, IPDI	XAS-4600	(see 3.1.1.2)

1/ Ammonium perchlorate: 31.0 percent by weight unground and 35.0 percent by weight fine ground.
2/ See 6.3.
3/ The iron oxide used shall be yellow iron oxide conforming to MIL-F-23938 with the modification that in table I of MIL-F-23938, "Iron as FE $De_2O_3.H_2O$" shall be substituted for "Iron as Fe_2O_3" and the calculation given for "Determination of Iron as Fe_2O_3" is not applicable.

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3.1.1.1 Ammonium perchlorate. The fine ground ammonium perchlorate shall have an average particle size of 4.2 ±.4 microns as determined by a Mine Safety Appliance (MSA) particle size analyzer. The ratio of fine ground to unground ammonium perchlorate may be adjusted to obtain the desired burning rate.

3.1.1.2 Isocyanate-hydroxyl ratio (NCO-OH). The liquid hydroxyl terminated polybutadiene (HTPB) and the propellant antioxidant compound are preblended at a 99:1 ratio. The equivalent weight (OH content) of the blend is determined or calculated. Then the weight percents of the HTPB and propellant antioxidant compound blend and the isophorone diisocyanate (IPDI) are adjusted to give an NCO-OH ratio of 0.75 to 0.90 for physical properties control. The total percent by weight of these ingredients must be equal to 16.000 (see table I).

3.1.1.3 Weighing accuracy. Each propellant ingredient shall be weighed to an accuracy of 0.3 percent of the value shown in table I or the nearest gram (gm), whichever is greater.

3.2 Chemical, physical, and mechanical properties. After curing 16 to 24 hours at 150 ±5 degrees Fahrenheit (° F) followed by curing at 135 ±5° F until a total cure time of 120 ± 2 hours has been completed, the propellant shall meet the requirements of table II.

TABLE II. Cured chemical, physical, and mechanical properties.

Property	Requirement	
	Maximum	Minimum
Density (lb/in^3)	0.0635	0.0625
Heat of explosion (cal/gm)	1,294	1,224
Burning rate at 77 ± 2° F (in/sec)		
1000 ± (lb/in^2)	0.790	0.680
2000 ± (lb/in^2)	1.180	1.020
3000 ± (lb/in^2)	1.480	1.320
Maximum stress at 77° F (lb/in^2)	-	120
Strain at maximum stress at 77° F (percent)	-	17.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein (see 6.2). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2). The Government

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reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lot size. For the purpose of inspection, the lot size of propellant shall consist of one manufacturer's propellant batch.

4.2.2 Sampling. Unless otherwise specified herein, two separate samples of fine ground ammonium perchlorate shall be taken during transfer from the mill collection system to the storage containers, and one sample of propellant shall be taken from each propellant batch. The samples shall be representative of the material from which they were withdrawn. The samples shall be placed in clean, dry containers and sealed after introduction of samples to preclude atmospheric contamination.

4.2.2.1 Ammonium perchlorate particle size determination. Two 9-milligram samples shall be taken from one of the fine ground ammonium perchlorate samples for testing to determine conformance to the requirements of 3.1.1.1. The fine ground ammonium perchlorate can be held in sealed containers 72 hours, maximum, at a temperature of 120 ±10° F before use. If held a greater length of time, the ammonium perchlorate shall be retested before use.

4.2.2.2 Chemical, physical, and mechanical properties determination. Samples shall be taken from propellant which has been vacuum cast into sample containers and cured 16 to 24 hours at 150 ±5° F followed by cure at 135 ±5° F until a total cure time of 120 ±2 hours has been completed. The samples shall be used to determine conformance to the requirements of 3.2.

4.2.3 Acceptance criteria. Unless otherwise specified herein, the criteria used for samples taken in accordance with 4.2.2 shall be as follows.

4.2.3.1 Ammonium perchlorate particle size. The sample shall be tested in duplicate. If the average particle size of the duplicate test results is within the limits of 3.1.1.1, the ammonium perchlorate will be acceptable.

4.2.3.2 Cured propellant density and heat of explosion. The propellant will be acceptable for density and heat of explosion if the average results of the two samples tested meet the applicable requirements of table II.

4.2.3.3 Cured propellant stress and strain. The propellant will be acceptable for cured propellant stress and strain if the average results of the five CPIA Class C specimens tested meet the applicable requirements of table II.

4.3 Test methods.

4.3.1 Average particle size. The average particle size of the fine ground ammonium perchlorate shall be determined by the MSA particle size analyzer, using

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the procedure recommended by the manufacturer of the instrument. Two 9-milligram samples of ammonium perchlorate shall be used.

4.3.2 Density. The density of the propellant shall be determined by immersion of a 15- to 25-gm sample into hexane, conforming to WS-7692 in accordance with the following procedure:

a. Assemble the required additions to the analytical balance as shown in ASTM D792, Section 7.
b. Cover the perforated glass basket with 0.5 to 0.75 inch of hexane. Maintain the same liquid level throughout any given series of determinations.
c. Add appropriate weights or counterweights to obtain a zero point.
d. Weigh the calibrated glass standard in air at "A" on the diagram, then in hexane at "B".
e. Calculate the density of the liquid as follows:

$$D_L = \frac{D_S \times (W_S - W_L)}{W_S}$$

where:

D_L = density of liquid, gm/cubic centimeter (cm^3)

D_S = density of glass standard, gm/cm^3

W_S = weight of glass standard in air, gm

W_L = weight of glass standard in liquid, gm.

f. Repeat procedure d. with the propellant sample in place of the glass standard.
g. Calculate the density of each sample as follows:

$$D_X = \frac{W_{XA} \times D_L}{W_{XA} - W_{XL}}$$

where:

D_X = density of sample, gm/cm^3

W_{XA} = weight of sample in air, gm

W_{XL} = weight of sample in liquid, gm

D_L = density of liquid, gm.

h. After completion of two density determinations, check the density of the hexane before proceeding to the next determination. Hexane densities must be determined to four significant figures.

i. Densities of all samples must be reported to four significant figures.

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4.3.3 Heat of explosion. Heat of explosion shall be determined in accordance with MIL-STD-286, Method 802.1, except that argon gas shall be used instead of nitrogen gas.

4.3.4 Burning rates. The propellant shall display the burning rates as described. Burning rates are determined by burning five propellant strands at the special starting pressure (±5 lb/in^2) and at 77 ± 2° F. The starting pressure and maximum pressure obtained shall be recorded and the average pressure determined. The strands shall be 0.250 ±0.025 inch by 0.125 ±0.025 inch or 0.250 ±0.250 inch in diameter by 5.5 ±0.5 inches long. Prior to burning, the strands shall be inhibited by dipping them into a mixture of 438 gm of methyl ethyl ketone, conforming to TT-M-261, and 253 gm of Acryloid Plastic B-7, or equivalent. Each strand shall be dipped three times with a minimum hold time of 20 minutes between dips. After the third dip, the strand shall be held a minimum of 60 minutes before further operations. Strand dipping shall be done at 77 ±5° F and 40 percent relative humidity or less. The length of a burned strand shall be 4.00 ±0.01 inches. A Crawford-type strand burning rate apparatus with an integral surge tank to prevent a rise in pressure of more than 50 pounds per square inch (lb/in^2) during burning shall be used.

4.3.5 Stress and strain. Stress and strain shall be determined in accordance with the CPIA Publication 21, Section 4.3.2.4, Class C, using a cross-head speed of 2 inches per minute. The samples shall be conditioned a minimum of 2 hours at test temperatures.

5. PACKAGING - Not applicable.

6. NOTES

6.1 Intended use. The propellant is intended for use in the rocket motor for the Rocket, High Explosive, 345 Milimeter: XM130.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Responsibility for inspection if different from 4.1.
c. Testing facilities if different from 4.1.

6.3 Aluminum oxide. Micro-Alumina Precision Lapping Powder (12-micron) manufactured by the K.C. Abrasive Company, Kansas City, Kansas 66115 has been found to perform satisfactorily in this formulation.

6.4 Propellant antioxidant compound. CAO-14 manufactured by the Ashland Chemical Company, 8 East Long St., Columbus, Ohio 43216 and AO-2246 manufactured by the American Cyanamid Company, Boundbrook, New Jersey 08805 have been found to perform satisfactorily in this formulation.

6.5 Acryloid Plastic B-7. A product manufactured by the Rohm and Haas Company, Independence Mall West, Philadelphia, Pennsylvania 19106. This product has been found to perform satisfactorily in testing this formulation.

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6.6 Sample calculation of binder ingredients.

Given: Laboratory analysis of HTPB, 0.74 milliequivalent per gram (meq/gm)
Theoretical composition of antioxidant.
5.75 meq/gm
Theoretical composition of IPDI, 9.0 meq/gm
NCO/OH = $\frac{0.9}{1.0}$

a. Find equivalent weight of HTPB/antioxidant mixture:

NOTE: Omit this calculation if a laboratory analysis of the HTPB/antioxidant has been performed to determine the meq or OH/gm.

A 100-gm mixture:

99 gm HTPB = 99 gm x 0.74 meq/gm = 73.26
meq 1 gm antioxidant = 1 gm x 5.75
meq/gm = 5.75 meq

$$\text{Total: } \frac{79.02 \text{ meq}}{100 \text{ gm HTPB/antioxidant}} = \frac{1 \text{ meq}}{1.266 \text{ gm}}$$

b. Find equivalent weight of IPDI:

$$\frac{9.0 \text{ meq}}{\text{gm IPDI}} = \frac{1 \text{ meq}}{0.111 \text{ gm IPDI}}$$

c. Use NCO/OH ratio:

$$\frac{\text{NCO}}{\text{OH}} = \frac{0.9 \text{ meq NCO}}{1.0 \text{ meq OH}} \quad \frac{0.9 \text{ meq } (0.111 \text{ gm/meq})}{1.0 \text{ meq } (1.266 \text{ gm/meq})}$$

$$= \frac{0.100 \text{ gm IPDI}}{1.266 \text{ gm HTPB/antioxidant}}$$

d. Convert to weight percent basis:

$$\text{HTPB/antioxidant} = \frac{1.261 \text{ gm HTPB/antioxidant}}{1.361 \text{ gm total}}$$

x 100 x 16% binder = 14.82% HTPB/antioxidant

$$\text{IPDI} = \frac{0.100 \text{ gm IPDI}}{1.361 \text{ gm total}} \times 100 \times 16\% \text{ binder} = 1.18\% \text{ IPDI}$$

6.7 Safety precautions. The loading, assembly and handling of the explosives, subassemblies and the finished items covered by this specification involve

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hazardous operations and, therefore, require suitable explosives safety precautions. Standard safety precautions for explosive-loaded items are dependent on the type of facility involved in the effort as follows:

Type of Activity	Governing Document
COCO (Contractor Owned-Contractor Operated)	DOD 4145.26 M, except that QD considerations are in Chapter 17 of DARCOM-R 385-100
GOGO Army (Govt Owned-Govt Operated)	DARCOM-R 385-100
GOCO Army (Govt Owned-Contractor Operated)	DARCOM-R 385-100
GOGO Navy	Ammunition Ashore, OP-5
GOCO Navy	Ammunition Ashore, OP-5

Note: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Shore, OP-5, should be made applicable.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 1340-A614

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